ΕΛΛΗΝΙΚΗ ΔΗΜΟΚΡΑΤΙΑ
ΠΑΝΕΠΙΣΤΗΜΙΟ ΜΑΚΕΔΟΝΙΑΣ

Τεχνικό Χρέος

Ποιότητα Λογισμικού

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Technical Debt Recap

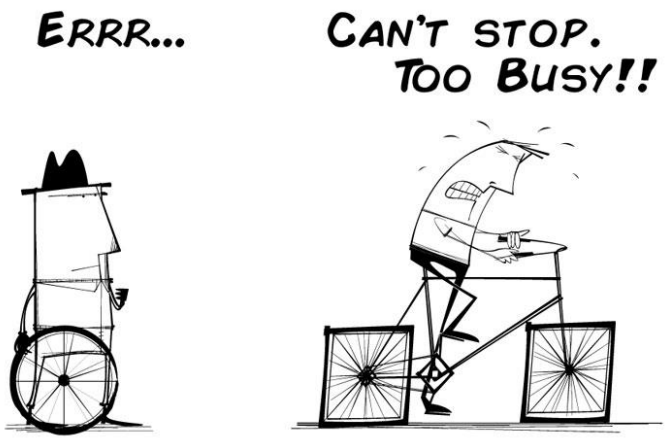
Technical Debt → The development of software at a lower quality that experiences increased maintenance costs

Short-term Benefits → Shorter time-to-market

Long-term Cost → Spend extra resources, when maintaining

TD Principal → Distance between a system's internal quality and the optimal quality level

TD Interest → Extra effort needed to maintain a system with accumulated technical debt

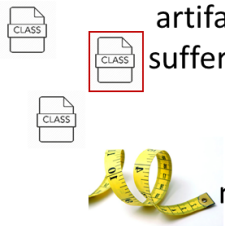


Technical Debt Recap

TD Dashboard

Visualizing TD

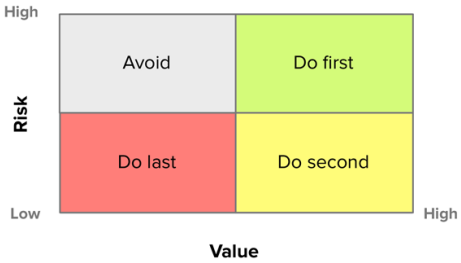
shows information that support:



artifacts that suffer from TD

measure their amount of interest and principal

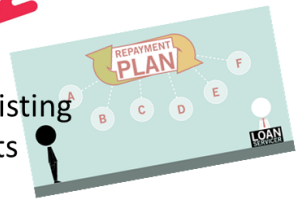
Quantifying TD



based on quantification

Prioritizing TD

prevent TD on new artifacts



refactor existing artifacts

Reducing TD

http://platform.sdk4ed.eu/

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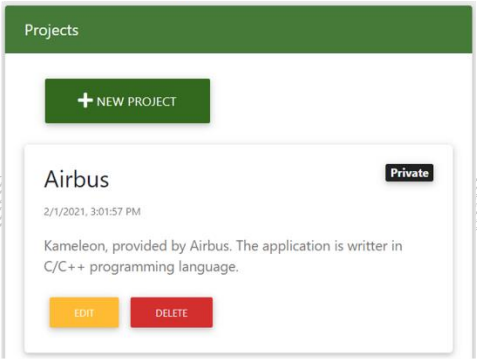
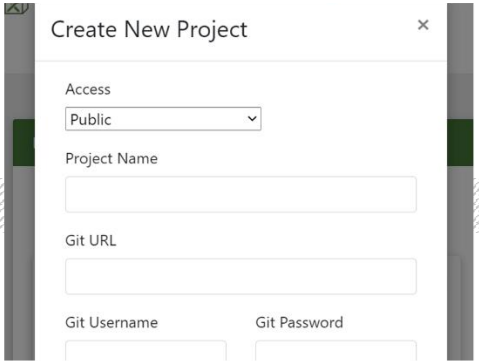
Project Selection

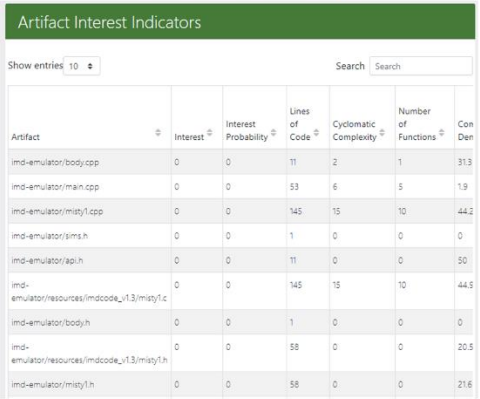


Create New Project

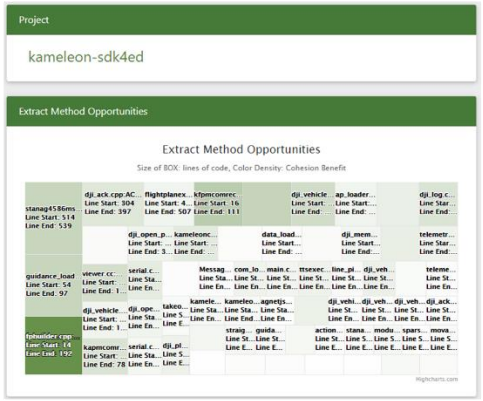


TD Evolution

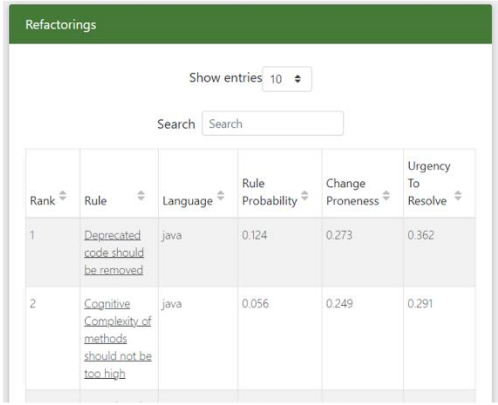
Artifact-Level Analysis



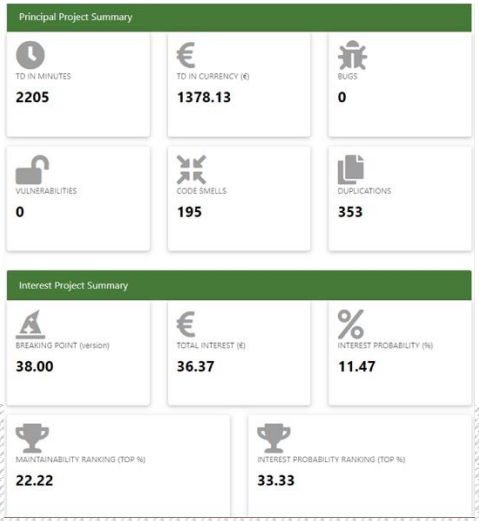
Refactoring Support



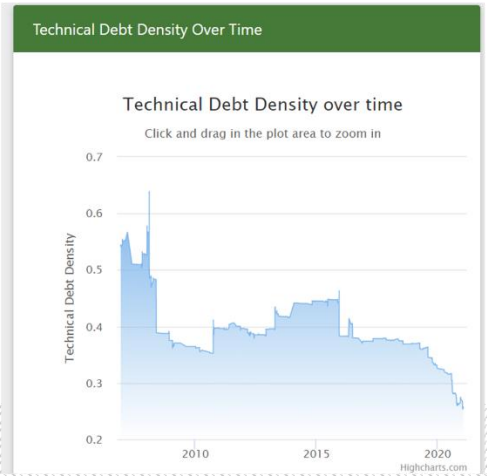
Common Quality Violations



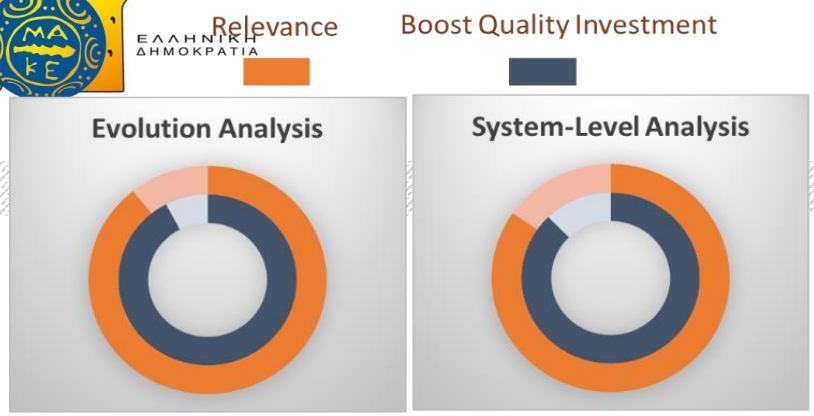
System-Level Analysis

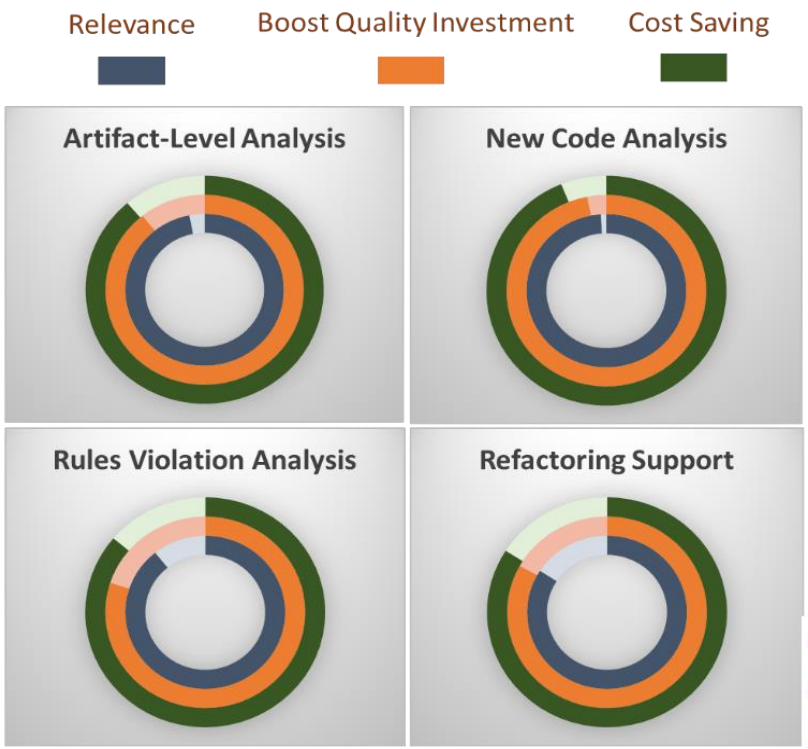


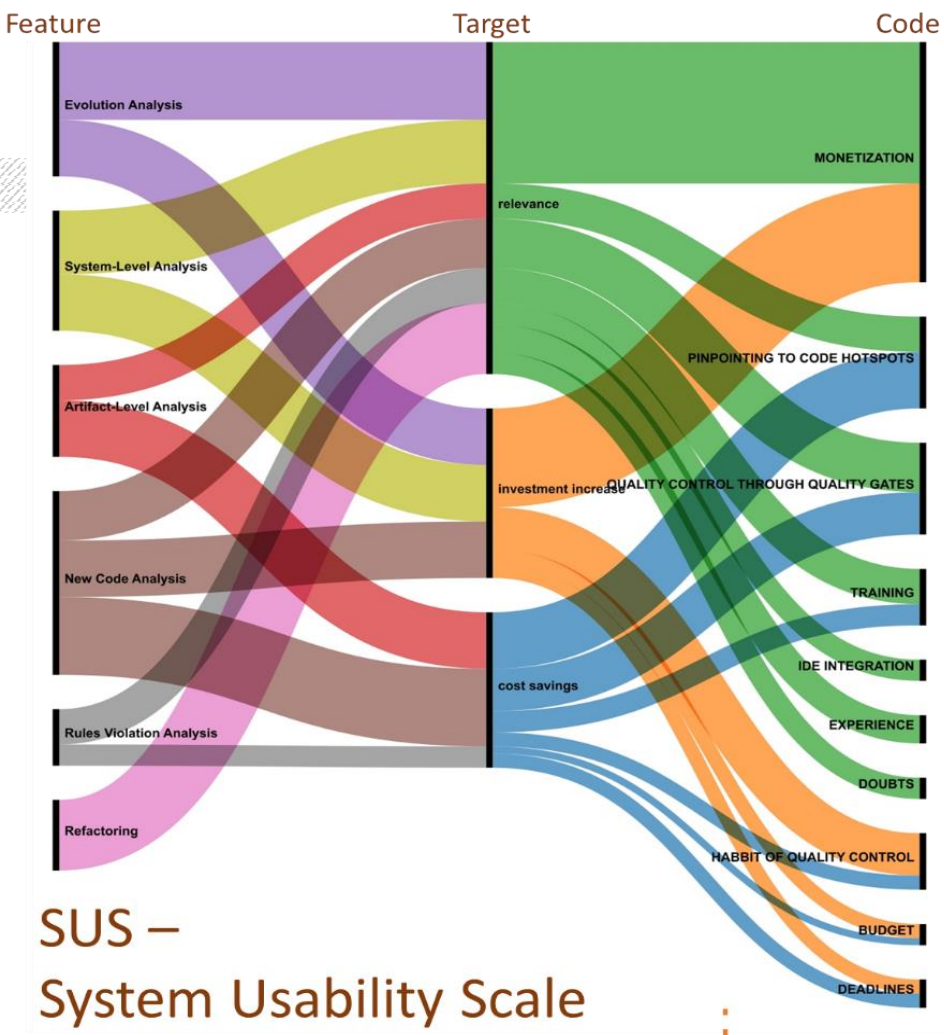
New Code Analysis



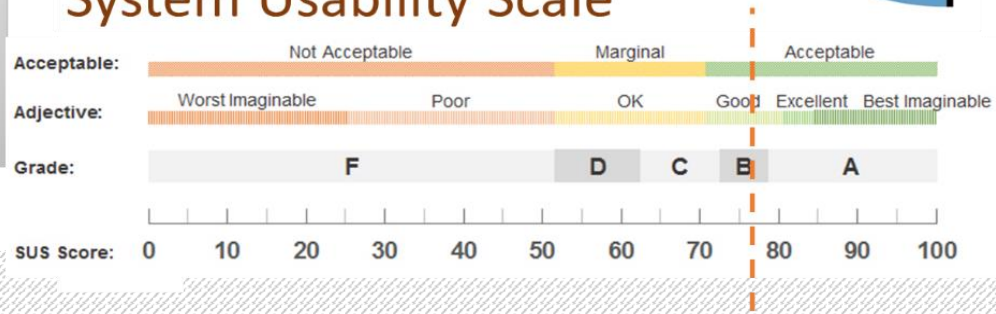
http://platform.sdk4ed.eu/







SUS – System Usability Scale



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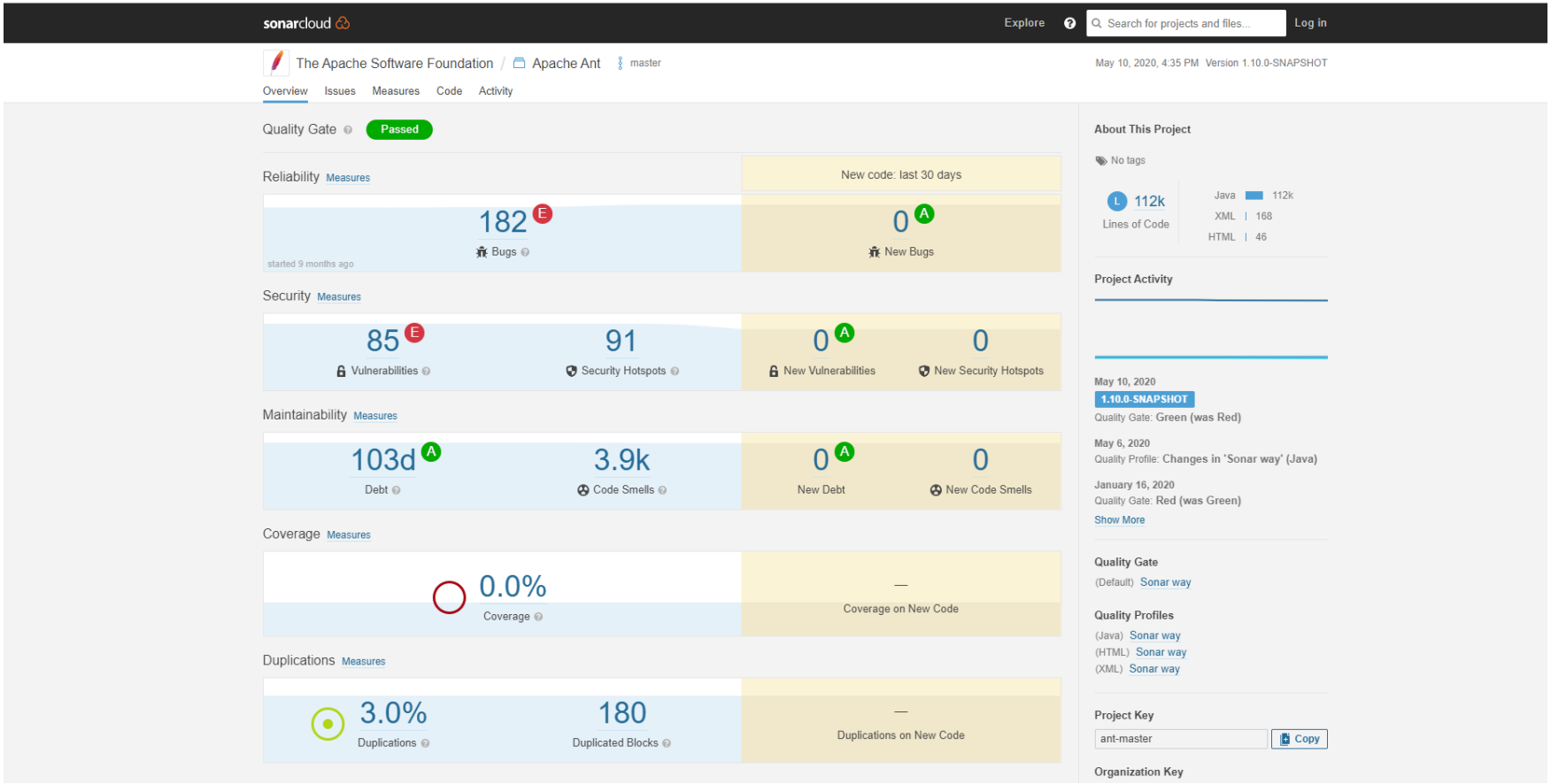
Principal Estimation

https://sonarcloud.io/organizations/apache/projects



Principal Estimation

https://github.com/tsantalis/JDeodorant

https://www.youtube.com/watch?v=h8K2M-lbDYo

https://github.com/tushartushar/DesigniteJava

$$TD_{Principal} = \sum_{i=0}^{types\ of\ smells} \#instances(i) \ * remediation\ time(i)$$

For remediation time you need to experiment

- apply 5 refactoring instances per type of smell
- use average time

Interest Estimation

- No Certain Methodology
- Uncertainty—Maintenance Frequency, Ease of Maintenance, Load of Maintenance

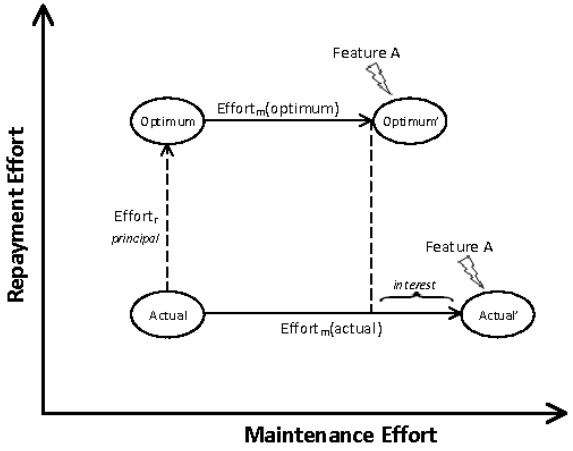
- **Maintainability** → A proxy to Ease of Maintenance

OO-Languages

- 10 maintainability indicators
- 5 quality properties related to maintainability in OO systems:
  (a) complexity
  (b) coupling
  (c) cohesion
  (d) Size
  (e) inheritance

Interest Estimation

- No Certain Methodology
- Uncertainty—Maintenance Frequency, Ease of Maintenance, Load of Maintenance



- So TD Interest for period (X) can be calculated as:

Distance $_{\text{from optimum maintainability}}$ * Maintenance Frequency * Period(X) * Load of Maintenance History

Interest Estimation

Maintainability Models

- sets of low-level metrics that quantify design-time quality attributes
- inheritance, cohesion, coupling, complexity, size
- somehow aggregated in one maintainability index

Accurate maintainability predictors

Metric	Description	Quality Attribute
DIT	Depth of Inheritance Tree: Inheritance level number, 0 for the root class.	*inheritance*
NOCC	Number of Children Classes: Number of direct sub-classes that the class has.	*inheritance*
MPC	Message Passing Coupling: Number of send statements defined in the class.	*coupling*
RFC	Response For a Class: Number of local methods plus the number of methods called by class methods.	*coupling*
LCOM	Lack of Cohesion of Methods: Number of disjoint sets of methods (a set of methods that do not interact with each other), in the class.	*cohesion*
DAC	Data Abstraction Coupling: Number of abstract types defined in the class.	*coupling*
WMPC	Weighted Method per Class: Average cyclomatic complexity of all methods in the class.	*complexity*
NOM	Number of Methods: Number of methods in the class.	*size*
SIZE1	Lines of Code: Number of semicolons in the class.	*size*
SIZE2	Number of Properties: Number of attributes and methods in the class	*size*

Interest Estimation

- Find similar classes
- Find the optimum value for each metric
- Calculate the difference between actual and optimal fitness values
- Assess the average distance between actual and optimal design
- Calculate interest

Similarity metrics

Metric	*Description*
Classes	Number of classes (including nested classes, interfaces, enums and annotations).
Complexity	The complexity calculated based on the number of paths through the code.
Functions	Number of functions.
NCLOC	Number of physical lines that contain at least one character which is neither a whitespace nor a tabulation nor part of a comment.
Sqale Index	Effort needed to fix all maintainability issues.
Statements	Number of statements.

- Number of LoC added in a unit of time
- Developer's hourly rate
  - ✓ **Interest in USD**

Interest Probability

http://www.cs.rug.nl/search/uploads/Resources/CommitChangeCalc.rar

http://www.cs.rug.nl/search/uploads/Resources/InstabilityCalculator.rar

$$\text{REM}_{\text{dependency}} = \frac{NDMC+NOP+NPrA}{NOM+NA} \quad (1)$$

NDMC: Number of distinct method calls from the dependent class to the source class (plus, super class method invocations for the case of generalization)

NOP: Number of polymorphic methods in the source class (valid only for generalization)

NPrA: Number of protected attributes in the source class (valid only for generalization or friend classes)

NOM: Number of methods in the source class

NA: Number of attributes in the source class

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Ερωτήσεις